

Enterprise Linux Server Installation & Configuration Guide

Course: ITEP 414 – System Administration and Maintenance

Institution: Laguna State Polytechnic University (LSPU)

System Administrator: gahi

Hostname: server01

1. Project Overview

This technical documentation covers the deployment, initial system configuration, network verification, and post-installation update routine of an Ubuntu Server instance hosted within an Oracle VirtualBox virtualization environment. The objective is to establish a secure, network-ready enterprise Linux server environment tailored for production workloads and system administration operations.

2. Objectives

Provision a virtual machine running a 64-bit Linux server operating system.

Configure administrative user access (gahi) and hostname identification (server01).

Verify network interface assignment (ip addr) and external Internet reachability (ping).

Update repositories and upgrade system packages to ensure security compliance.

Document core architectural concepts including firmware modes (BIOS vs. UEFI) and the Linux boot sequence.

3. Software Requirements

Hypervisor: Oracle VM VirtualBox 7.x

Operating System ISO: Ubuntu Server (64-bit Architecture, Linux Kernel 7.0.x / Ubuntu Server LTS)

Terminal Emulator / Console: VirtualBox TTY Virtual Console

4. Virtual Machine Specifications

Parameter	Allocated Specification
VM Name	Ubuntu-Server-Week03
Architecture	x86_64 (64-bit)
Assigned Hostname	server01
Primary User Account	gahi
vCPU Allocation	1–2 vCPUs
Base Memory (RAM)	2048 MB (2 GB)
Virtual Hard Disk Size	20.00 GB (18.53 GB usable partition)
Network Adapter	NAT / Bridged (enp0s3)

5. Step-by-Step Installation Procedure

Step 1: Virtual Machine Provisioning

Launch Oracle VM VirtualBox and select New.

Set the OS Type to Linux and Version to Ubuntu (64-bit).

Attach the downloaded Ubuntu Server ISO image to the virtual optical drive.

Complete the automated/guided installer steps:

Choose installation language and keyboard layout.

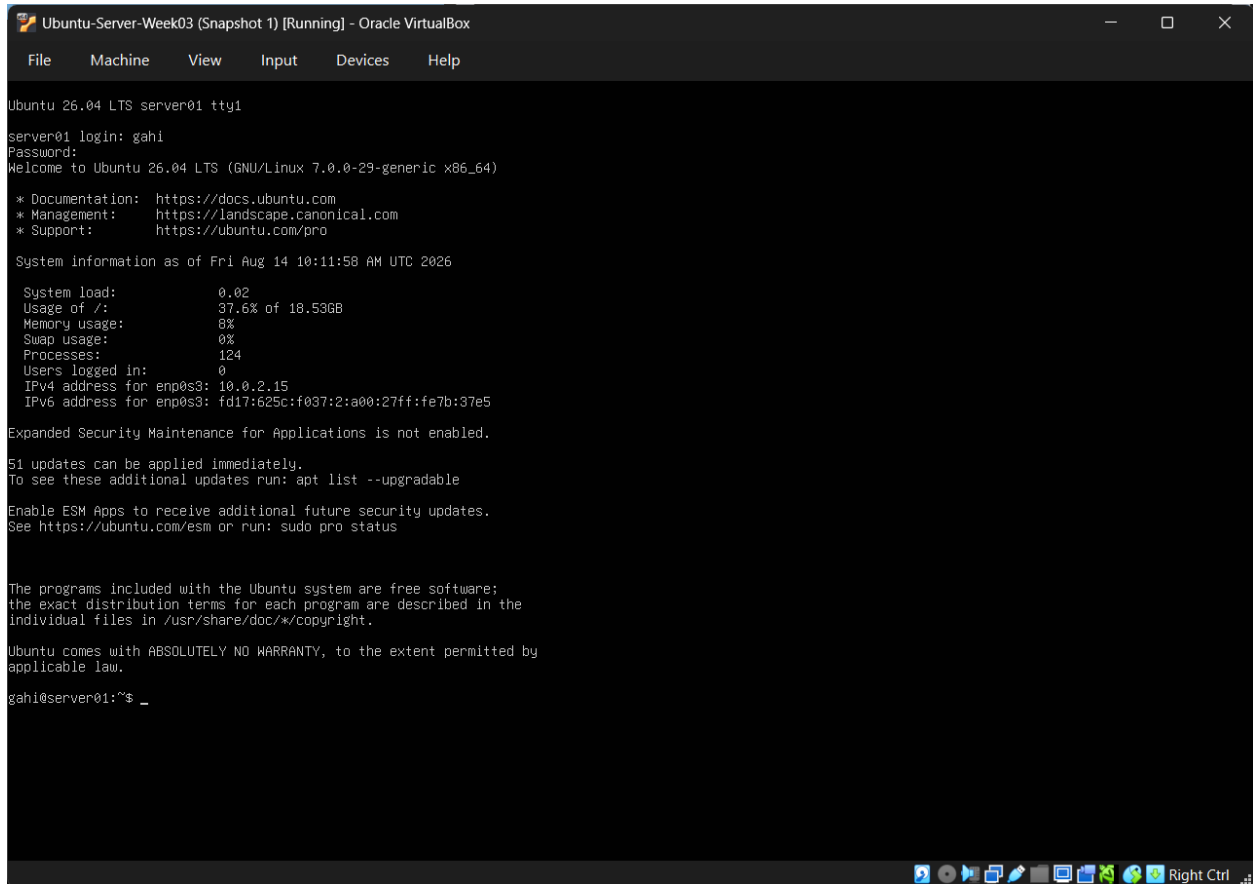
Configure basic network addressing via DHCP on interface enp0s3.

Partition the disk using guided entire disk layout (ext4 root filesystem).

Create the administrative profile: Username: gahi | Server Name: server01.

Step 2: First Boot & User Authentication

Log into the server terminal using the provisioned credentials:



```
Ubuntu 26.04 LTS server01 tty1
server01 login: gahi
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Aug 14 10:11:58 AM UTC 2026

System load:          0.02
Usage of /:            37.6% of 18.53GB
Memory usage:         8%
Swap usage:           0%
Processes:            124
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe7b:37e5

Expanded Security Maintenance for Applications is not enabled.

51 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

gahi@server01:~$ _
```

Bash

Ubuntu 26.04 LTS server01 tty1

server01 login: gahi

Password: [entered]

Welcome to Ubuntu (GNU/Linux 7.0.0-29-generic x86_64)

6. Configuration Details & Verification Commands

Verification 1: Hostname Resolution

Verify that the system hostname is correctly recognized across local shell environments.

Bash

hostname

Output: server01

```
Ubuntu 26.04 LTS server01 tty1
server01 login: gahi
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
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System load:          0.02
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Memory usage:         8%
Swap usage:           0%
Processes:            124
Users logged in:      0
IPV4 address for enp0s3: 10.0.2.15
IPV6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe7b:37e5

Expanded Security Maintenance for Applications is not enabled.

51 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

gahi@server01:~$ hostname
server01
gahi@server01:~$ hostname hostname
```

Verification 2: Network Interface & IP Assignment

Check the state of local loopback (lo) and primary network adapter (enp0s3).

Bash

ip addr

Loopback (lo): 127.0.0.1/8

Ethernet (enp0s3): 10.0.2.15/24 (MAC: 08:00:27:7b:37:e5, State: UP)

```

Usage: hostname [-b] {hostname|-F file}      set host name (from file)
hostname [-a|-A|-d|-f|-i|-I|-s|-y]        display formatted name
hostname                                     display host name

        {yp,nis,}domainname {nisdomain|-F file} set NIS domain name (from file)
        {yp,nis,}domainname                display NIS domain name

        dnsdomainname                       display dns domain name

        hostname -V|--version|-h|--help     print info and exit

Program name:
        {yp,nis,}domainname=hostname -y
        dnsdomainname=hostname -d

Program options:
        -a, --alias                alias names
        -A, --all-fqdns            all long host names (FQDNs)
        -b, --boot                 set default hostname if none available
        -d, --domain               DNS domain name
        -f, --fqdn, --long         long host name (FQDN)
        -F, --file                 read host name or NIS domain name from given file
        -i, --ip-address           addresses for the host name
        -I, --all-ip-addresses     all addresses for the host
        -s, --short                short host name
        -y, --yp, --nis            NIS/YP domain name

Description:
This command can get or set the host name or the NIS domain name. You can
also get the DNS domain or the FQDN (fully qualified domain name).
Unless you are using bind or NIS for host lookups you can change the
FQDN (Fully Qualified Domain Name) and the DNS domain name (which is
part of the FQDN) in the /etc/hosts file.
gahi@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:7b:37:e5 brd ff:ff:ff:ff:ff:ff
    altname enx0800277b37e5
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85624sec preferred_lft 85624sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7b:37e5/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86240sec preferred_lft 14240sec
    inet6 fe80::a00:27ff:fe7b:37e5/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
gahi@server01:~$

```

Verification 3: Connectivity & DNS Resolution

Verify external network route connectivity and ICMP transmission by pinging an external host.

Bash

ping -c 4 google.com

Result: 4 packets transmitted, 4 received, 0% packet loss, time 3045ms (rtt min/avg/max = 8.282/10.263/11.221 ms).

```

Program name:
  {yp,nis,}domainname=hostname -y
  dnsdomainname=hostname -d

Program options:
  -a, --alias            alias names
  -A, --all-fqdns        all long host names (FQDNs)
  -b, --boot            set default hostname if none available
  -d, --domain          DNS domain name
  -f, --fqdn, --long    long host name (FQDN)
  -F, --file            read host name or NIS domain name from given file
  -l, --ip-address      addresses for the host name
  -I, --all-ip-addresses all addresses for the host
  -s, --short           short host name
  -y, --yp, --nis       NIS/YP domain name

Description:
  This command can get or set the host name or the NIS domain name. You can
  also get the DNS domain or the FQDN (fully qualified domain name).
  Unless you are using bind or NIS for host lookups you can change the
  FQDN (Fully Qualified Domain Name) and the DNS domain name (which is
  part of the FQDN) in the /etc/hosts file.

gahi@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
   link/ether 08:00:27:7b:37:e5 brd ff:ff:ff:ff:ff:ff
   altname enx0800277b37e5
   inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 85624sec preferred_lft 85624sec
   inet6 fd17:625c:f037:2:a00:27ff:fe7b:37e5/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86240sec preferred_lft 14240sec
   inet6 fe80::a00:27ff:fe7b:37e5/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever

gahi@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.238) 56(84) bytes of data:
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=1 ttl=64 time=11.2 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=2 ttl=64 time=11.2 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=3 ttl=64 time=8.28 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=4 ttl=64 time=10.3 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3045ms
rtt min/avg/max/mdev = 8.282/10.263/11.221/1.200 ms
gahi@server01:~$

```

Verification 4: Package Management & System Upgrade

Synchronize package indexes and execute distribution updates:

Bash

`sudo apt update && sudo apt upgrade -y`

Output Details: Processed triggers for systemd, rsyslog, dbus, and regenerated initial RAM disk (update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic).

```
Setting up bind9-host (1:9.20.24-1ubuntu0.1)...
Setting up linux-firmware-misc (20260319.git217ca6e4-0ubuntu2.2)...
Setting up linux-firmware-qualcomm-graphics (20260319.git217ca6e4-0ubuntu1.1)...
Setting up fwupd (2.1.1-1ubuntu3.1)...
fwupd-refresh.service is a disabled or a static unit not running, not starting it.
fwupd.service is a disabled or a static unit not running, not starting it.
Setting up sg3-utils (1.48-3ubuntu3.1)...
Setting up cloud-init (26.1-0ubuntu3~26.04.1)...
Setting up ubuntu-minimal (1.570.2)...
Setting up sg3-utils-udev (1.48-3ubuntu3.1)...
update-initramfs: deferring update (trigger activated)
Setting up bind9-dnsutils (1:9.20.24-1ubuntu0.1)...
Setting up ubuntu-standard (1.570.2)...
Setting up apport-core-dump-handler (2.34.1-0ubuntu0.1)...
Setting up apport (2.34.1-0ubuntu0.1)...
apport-autoreport.service is a disabled or a static unit not running, not starting it.
Setting up ubuntu-server (1.570.2)...
Setting up ubuntu-server-minimal (1.570.2)...
Processing triggers for plymouth (24.004.60+git20250831.4a3c171d-0ubuntu0)...
Processing triggers for libc-bin (2.40-2ubuntu2.3)...
Processing triggers for rsyslog (8.2512.0-1ubuntu4.1)...
Processing triggers for systemd (259.5-0ubuntu3.4)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for dbus (1.16.2-2ubuntu4)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
^[[    Processing triggers for procps (2:4.0.4-9ubuntu1)...
Processing triggers for install-info (7.2-5ubuntu2)...
Processing triggers for plymouth-theme-ubuntu-text (24.004.60+git20250831.4a3c171d-0ubuntu0)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-29-generic
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service

No containers need to be restarted.

User sessions running outdated binaries:
gahi @ user manager: (sd-pam) [1467]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
gahi@server01:~$
```

7. BIOS vs. UEFI Comparison

Criteria	BIOS (Basic Input/Output System)	UEFI (Unified Extensible Firmware Interface)
Definition	Legacy firmware stored on a ROM/flash chip that initializes hardware and loads the OS.	Modern firmware interface offering a flexible environment and advanced hardware control.
Boot Process	Executes POST, reads MBR on Sector 0, hands off to the bootloader.	Reads EFI System Partition (ESP) directly and executes .efi bootloader binaries.
Max Disk Support	2.2 TB drive capacity limit (32-bit LBA).	Up to 9.4 ZB (Zettabytes) with 64-bit addressing.
Partition Style	Master Boot Record (MBR) — maximum 4 primary partitions.	GUID Partition Table (GPT) — up to 128 standard partitions.

Criteria	BIOS (Basic Input/Output System)	UEFI (Unified Extensible Firmware Interface)
Security	Basic BIOS passwords; vulnerable to bootkits.	Secure Boot with cryptographic signature enforcement.
Speed	Slower sequential hardware checks.	Faster parallel hardware initialization.

8. Boot Process Flowchart

Code snippet



9. Windows Server Installation Summary

Installation Media: Windows Server 2022/2025 Standard ISO.

Setup Flow: Select Locale $\rightarrow$ Choose Desktop Experience or Server Core $\rightarrow$ Accept License $\rightarrow$ Custom Drive Allocation $\rightarrow$ Set Local Administrator Password.

Initial Tasks: Rename computer name, assign static IP configuration, and enable Remote Desktop (RDP) via Server Manager.

10. Operating System Comparison

Feature	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial Proprietary (Per-core + CALs)	Free & Open Source (GPL)	Free & Open Source (RHEL-compatible)
Default UI	Graphical Desktop Experience	Headless CLI (Terminal)	Headless CLI (Cockpit Web UI)
Package Tool	Windows Update / MSI / Winget	APT (.deb) / Snap	DNF / YUM (.rpm)
Access Control	Active Directory / GPO	AppArmor / UFW	SELinux / Firewalld
Primary Use	Domain Controller, Exchange, File Share	Cloud Native, Web Hosting, Docker	Enterprise Databases, High- Load Servers

11. Troubleshooting & Lessons Learned

Problems Encountered

Command Syntax Misalignment: Inadvertently executing `hostname hostname` instead of running `hostname` with query flags, which triggered the utility's help output.

Post-Upgrade Service Deferrals: During `apt upgrade`, `needrestart` deferred restarting specific background daemons (`dbus.service`) while active user sessions remained open.

Solutions Implemented

Clarified syntax differences between inspecting (`hostname`) and writing host configurations (`hostnamectl set-hostname`).

Monitored daemon triggers and allowed automated `update-initramfs` script sequences to finalize kernel updates before rebooting.

Lessons Learned

Pre-Flight Planning: Validating network configurations (ip addr and ping) immediately after operating system provisioning prevents downstream software installation failures.

Kernel & Initramfs Awareness: Observing how apt upgrade invokes update-initramfs emphasized the role of initial RAM disks in connecting the Linux kernel to root block devices during startup.

12. References

Canonical Ltd. (2024). Ubuntu Server Guide & Documentation. Retrieved from <https://docs.ubuntu.com>

Oracle Corporation. (2023). Oracle VM VirtualBox User Manual. Retrieved from <https://www.virtualbox.org/manual/>

Linux Foundation. (2023). Systemd System and Service Manager Overview. Retrieved from <https://systemd.io/>